



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	This morning, the organization experienced a Distributed Denial of Service (DDoS) attack that lasted for two hours. Malicious actors exploited an unconfigured firewall to flood the internal network with a high volume of ICMP packets . This attack overwhelmed network resources, making all internal network services and resources unavailable to employees. The incident management team responded by blocking the incoming ICMP traffic, temporarily taking non-critical services offline, and restoring critical network services to resolve the immediate incident.
Identify	The incident team identified the event as a DDoS attack . The vulnerability exploited was an unconfigured firewall that failed to filter or limit malicious traffic. The attack vector was a flood of ICMP pings . The affected systems were the entire internal network and all network services , which were inaccessible to internal users during the attack.
Protect	To protect assets from a similar attack, the network security team implemented several new safeguards. These include:

	* A new firewall rule to limit the rate of incoming ICMP packets. * Source IP address verification on the firewall to check for and block spoofed IP addresses. * An Intrusion Detection/Prevention System (IDS/IPS) to help filter ICMP traffic based on suspicious characteristics.
Detect	To improve the detection of future incidents, the organization has implemented network monitoring software. This software is configured to analyze traffic patterns and detect anomalies, such as a sudden spike in ICMP traffic, and send alerts to the security team. The new IDS/IPS will also serve as a detection tool by logging and alerting on potential attack signatures.
Respond	A formal incident response plan for DDoS attacks will be developed. This plan will include: * Containment: Procedures for security staff to quickly apply firewall rules to block or rate-limit the identified attack traffic. * Mitigation: A process for triaging services and taking non-critical systems offline to preserve bandwidth for critical operations.

	* Communication: A clear plan for notifying employees about the service disruption and providing status updates.
Recover	The recovery plan prioritizes business continuity by focusing on a phased restoration of services. * Immediate Recovery: First priority is to restore all critical network services as soon as the attack is contained. * Full Recovery: Once the network is stable, all non-critical services that were taken offline will be systematically brought back online and tested. * Communication: The IT team will communicate to all staff once services are fully restored.

Reflections/Notes: The incident highlighted a critical vulnerability in the firewall configuration. The implemented solutions (rate limiting, IP verification, IDS/IPS, and monitoring software) directly address the technical gaps. Future work should include regular firewall rule and configuration audits to prevent similar oversights.